Initials:_____

BME 105: Genetics in the Genomics Era
Spring 2022: Midterm 2
Extended Calculation Problem

Name: _____

Student ID:_____

Before starting, write your initials on the top of each page.

You can use the back-side of the pages for scratch, but we will not grade answers written on the back-side of the page.

THIS IS ONLY THE EXTENDED CALCULATION QUESTION. THE OTHER PART OF YOUR EXAM IS ON CANVAS.

PLEASE CHOOSE ONLY ONE BELOW AND SUBMIT TO PROCTOR AT THE END OF THE EXAM:

- ☐ **My answers are submitted as physical paper (this packet)**
- ☐ **My answers were submitted electronically.**

If submitted electronically, you only need to return the top sheet of this packet.

Section	Points
Part IV: Extended calculation	/55

Second Letter

1st letter	U		C		A		G		3rd letter
U	UUU UUC UUA UUG	Phe Leu	UCU UCC UCA UCG	Ser	UAU UAC UAA UAG	Tyr Stop Stop	UGU UGC UGA UGG	Cys Stop Trp	U C A G
C	CUU CUC CUA CUG	Leu	CCU CCC CCA CCG	Pro	CAU CAC CAA CAG	His Gln	CGU CGC CGA CGG	Arg	U C A G
A	AUU AUC AUA AUG	Ile Met	ACU ACC ACA ACG	Thr	AAU AAC AAA AAG	Asn Lys	AGU AGC AGA AGG	Ser Arg	U C A G
G	GUU GUC GUA GUG	Val	GCU GCC GCA GCG	Ala	GAU GAC GAA GAG	Asp Glu	GGU GGC GGA GGG	Gly	U C A G

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Part IV: Extended calculations (show your work for partial credit, show pedigrees and genotypes, etc.).

1. A plant breeding true for purple flowers was crossed with a plant true breeding for white flowers. The F1 progeny all had purple flowers. When the F1 were crossed among themselves, they produced 81 plants with purple flowers and 63 plants with white flowered plants. When the F1 plants were backcrossed to the white-flowered parent, 78 plants had white flowers and 24 had purple flowers. Coloration in plants is often controlled by more than one gene.

   a. What are the genotypes and corresponding phenotypes of parental, F1, and F2 plants? You can use any letters you want for genes/alleles but use capital letters for dominant alleles and lowercase for recessive alleles. Show your work and explain how you figured out the genotypes for each phenotype (**15 points**).

F2 81 purple : 63 white

9 : 7 ratio

Suggests complementary/ reciprocal recessive epistasis

P AABB × aabb

F1 AaBb

F2 A-B- purple

A-bb, aaB-, aabb } white

Check backcross!

F1 AaBb × P aabb

A-Bb → each → 1/4 purple

aaB- → 1/4 prob.

A-bb →

aabb →

} 3/4 white

≈ 78 white : 24 purple

3 : 1

   b. What type of genetic interaction is there, if any (i.e. what type of epistasis, if any)? (**5 points**)

reciprocal recessive epistasis

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2. In a GWAS study to find breast cancer susceptibility loci, 580,000 SNPs were genotyped among 350 cases of breast cancer and 500 controls using SNP microarrays. At one SNP that is either C or A, they found that 38% of cases had a C while 28% of controls had a C.
   a. Using a chi-square test, determine if there is a significant association of this SNP with breast cancer. Show your work. Include the chi-square statistic, the degrees of freedom and the p-value. Given your p-value, does this SNP have a significant association with breast cancer? (**15 points**)

	Cases	Controls	
C	O = 350 × .38 = 133 E = 273/850 × 350 = 112 X^2 = (112 − 133)²/112 3.9	O = 500 × .28 = 140 E = 273 − 112 = 161 X^2 = 2.7	273
A	O = 350 − 133 = 217 E = 350 − 112 = 238 X^2 = 1.9	O = 500 − 140 = 360 E = 500 − 161 = 339 X^2 = 1.3	577
	350	500	850

X^2 = 3.9 + 2.7 + 1.9 + 1.3 = 9.8 df

chidist(9.8, 1)

P = 0.001745

not significant

threshold is 0.05/580,000

   b. This SNP is located in an enhancer of the gene *MAP3K1* and was the most significant associated SNP in the GWAS study. It is known that this enhancer SNP is in linkage disequilibrium with a missense variant of the *MAP3K1* gene. The function of the missense variant is currently unknown. Which SNP is most likely the causal SNP for breast cancer susceptibility and why? (**5 points**)

Many possible answers. However, the most correct answer is that you cannot know which is the causal SNP until you do further experiments

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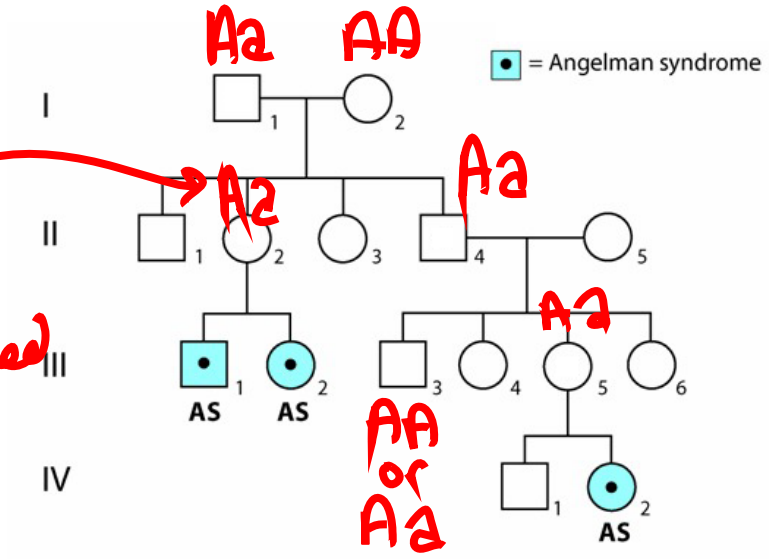
3. Below is a pedigree of the inheritance of Angelman syndrome. This is a disease that is due to the expression of a rare recessive allele of an autosomal gene that is paternally imprinted. Here, **A** is the normal dominant allele, and **a** is the recessive disease allele.

has to be a carrier but since does not have Angelman, must have received allele from father and it is imprinted



a. Indicate all possible genotypes for I-1, I-2, II-2, II-4, III-3, III-5 (**12 points**).

b. Is this an example of complete penetrance, incomplete penetrance, unvarying expressivity, and/or variable expressivity and why? (**3 points**)

incomplete penetrance → same genotype different phenotype

unvarying expressivity → no variation in phenotype shown in pedigree.